

Project: Part 1 (5pts) due Wed Oct 29, Part 2 (5pts) due Wed Nov 19, 11:59 pm.

1. **Coherent States.** [Numerical, TISE code] Consider a spinless particle with mass $m = 1$ in 1D, subject to the potential $V(x) = x^2/2$.

Part 1: (a) Plot this potential in the interval $x \in [-R, R]$, with $R = 10$, and describe your qualitative expectations for the spectrum and the eigenstates of the Hamiltonian. (b) Use the TISE program to compute the eigenstates and the eigenvalues of the Hamiltonian, for the parameters in point (a), imposing zero boundary conditions. (c) Prepare a two-slides presentation outlining the main features of this system. The slides must include a plot of the potential superposed to the energy levels you found (up to $E = 11$), as well as a plot of the first four eigenstates of the Hamiltonian, ϕ_n , $n = 0, 1, 2, 3$ (one plot per eigenfunction).

Part 2: Simulate the motion of a wavepacket $\psi(x, t)$, initially in the coherent state $\psi(x, 0) = e^{ik_0 x} \phi_0(x - x_0)$, by expanding it in the eigenstates of the Harmonic oscillator you have determined previously. Consider three cases: i) $x_0 > 0$ and $k_0 = 0$; ii) $x_0 = 0$ and $k_0 > 0$; iii) $x_0 > 0$, $k_0 > 0$. For each of these cases, plot the square module of the wave function computed at different times as

$$\psi(x, t) = \sum_{n=0}^N \varphi_n(x) c_n(0) e^{-iE_n t/\hbar},$$

and compare the result with the analytical predictions based on the theory of coherent states that we have studied in class. For the numerical details, see the instructions for part 2 of proj. #3.

2. **Ammonia inversion.** [Numerical, TISE code] Consider a spinless particle with mass $m = 1$ in 1D, subject to the following potential,

$$V(x) = \alpha x^4 - \frac{1}{4}x^2 + \frac{1}{64\alpha}, \quad \alpha > 0.$$

Part 1: (a) Plot this potential for $\alpha = 1/200$, in the interval $x \in [-R, R]$, with $R = 10$, and describe your qualitative expectations for the spectrum and the eigenstates of the Hamiltonian. (b) Use the TISE program to compute the eigenstates and eigenvalues of the Hamiltonian, for the parameters in point (a), imposing zero boundary conditions. (c) Prepare a two-slides presentation outlining the main features of this system. The slides must include a plot of the potential superposed to the energy levels you found (up to $E < V(0)$), as well as a plot of the first four eigenstates of the Hamiltonian (one plot per eigenfunction).

Part 2: Determine the evolution of the wave function in the following four cases: i) $\psi(t=0) = (\phi_1 + \phi_2)/\sqrt{2}$; ii) $\psi(t=0) = (\phi_3 + \phi_4)/\sqrt{2}$; iii) $\psi(t=0) = (\phi_5 + \phi_6)/\sqrt{2}$; iv) $\psi(t=0) = (\phi_1 + \phi_2 + \phi_3 + \phi_4)/2$ (or the necessary sign needed to localize the initial state in one of the two wells of the potential). Plot the probability density for the particle in each of these four cases for a number of times, and determine the time necessary for the state to transition to the opposite side of the potential barrier. Using the results for the first three cases, plot the logarithm of the transition time as a function of the height of the potential barrier with respect to the expectation value of the energy of the initial state. What conclusions can you infer from this study?

3. **Morse Potential.** [Numerical, TISE code] Consider a spinless particle with mass $m = 100$ in 1D, subject to the following potential,

$$V(x) = \begin{cases} D_e [1 - e^{-a(x-x_e)}]^2 & x \geq 0 \\ \infty & x < 0 \end{cases}$$

Part 1: (a) Plot this potential for $D_e = 1$, $a = 1$, $x_e = 1$, in the interval $x \in [0, R]$, with $R = 10$, and describe your qualitative expectations for the spectrum and the eigenstates of the Hamiltonian. (b) Use the TISE program to compute the bound eigenstates and the eigenvalues of the Hamiltonian, for the parameters in point (a), imposing zero boundary conditions. (c) Prepare a two-slides presentation outlining the main features of this system. The slides must include a plot of the potential superposed to the energy levels you found (up to $E = D_e$), as well as a plot of the first four eigenfunctions of the Hamiltonian (one plot per eigenfunction).

Part 2: Simulate the dynamics of a coherent superposition of these states. To do so, compute the expansion coefficients, in the basis of the eigenstates of the Morse potential, of a Gaussian state with the same parameters as the ground state of the harmonic potential tangent to the Morse potential at its minimum, but centered at an arbitrary prescribed point $x_0 \neq x_e$. As long as you encode correctly the Gaussian function, the calculation of the expansion coefficients is rather straightforward. Let's indicate an individual eigenstate of the Morse potential with $|\varphi_i\rangle$, $\hat{H}|\varphi_i\rangle = |\varphi_i\rangle E_i$. These states are expanded in B-splines, whose coefficients are returned by the diagonalization routine, $|\varphi_i\rangle = \sum_j |B_j\rangle C_{ji}$. We can be more concise by defining a row vector of kets as $|\varphi\rangle = (|\varphi_1\rangle, |\varphi_2\rangle, \dots, |\varphi_n\rangle)$ and $|\mathbf{B}\rangle = (|B_1\rangle, |B_2\rangle, \dots, |B_n\rangle)$, in which case the expansion becomes $|\varphi\rangle = |\mathbf{B}\rangle \mathbf{C}$. The expansion coefficients of a shifted Gaussian $|G\rangle$, then, are given by

$$\langle\varphi|G\rangle = \mathbf{C}^\dagger \langle\mathbf{B}|G\rangle, \quad \langle B_i|G\rangle = \int dx B_i(x) G(x).$$

The B-spline module available to you does not have these integrals implemented. However, it is easy to reconstruct them thanks to the fact that $\sum_i B_i(x) = 1$, and hence

$$\langle B_i|G\rangle = \sum_j \int dx B_i(x) G(x) B_j(x)$$

which is just the matrix element of the "potential" $G(x)$ evaluated between a pair of B-splines, and summed over one of the two B-spline indexes. This integral, of course, is readily implemented. Once you have the coefficients, you can evaluate the dynamics of the wavepacket as

$$|\psi(t)\rangle = e^{-i\hat{H}t/\hbar} |G\rangle = |\varphi\rangle e^{-i\mathbf{E}t/\hbar} \langle\varphi|G\rangle = |\mathbf{B}\rangle \mathbf{C} e^{-i\mathbf{E}t/\hbar} \mathbf{C}^\dagger \langle\mathbf{B}|G\rangle$$

To compute the expansion coefficients of the wave packet as a function of time on the B-spline basis (which we'll need to plot the wavepacket!), you need to take two preliminary steps:

- (a) compute $\langle B_i|G\rangle = \sum_j \int dx B_i(x) G(x) B_j(x)$
- (b) compute $\langle\varphi|G\rangle = \mathbf{C}^\dagger \langle\mathbf{B}|G\rangle$ (matrix-vector multiplication)

and then, for all the times of interest (you'd do a loop), compute the following

- (a) compute $\mathbf{V}(t) = e^{-i\mathbf{E}t/\hbar} \langle\varphi|G\rangle$ (diagonal matrix - vector multiplication)
- (b) compute $\mathbf{C}\mathbf{V}(t)$ (matrix - vector multiplication)

Finally, you tabulate the wavefunction and print it to file, and plot the wavefunction as a function of time. You should repeat this procedure for various values of x_0 , to illustrate the deviation from a Harmonic oscillator.

4. **Predissociation.** [Numerical, TISE code] Given the following functions, for $x > 0$

$$E_{\text{Morse}}(x) = D_e \left[1 - e^{-a(x-x_e)} \right]^2, \quad E_{\text{Coulomb}}(x) = \frac{1}{x} + \Delta, \quad E_{\text{CI}}(x) = \alpha \frac{x}{x^2 + 1},$$

compute the potentials $V_+(x)$ and $V_-(x)$, defined as, respectively, the smallest and largest eigenvalues of the matrix

$$\mathbf{V}(x) = \begin{pmatrix} E_{\text{mor}}(x) & E_{\text{CI}}(x) \\ E_{\text{CI}}(x) & E_{\text{cou}}(x) \end{pmatrix}.$$

Part 1: (a) Plot the potentials $E_{\text{Morse}}(x)$, $E_{\text{Coulomb}}(x)$, and $V_{\pm}(x)$ for $D_e = 1$, $a = 1$, $x_e = 1$, $\Delta = 0.5$, and $\alpha = 0.3$ in the interval $x \in [0, R]$, with $R = 20$. Describe qualitatively your expectation for the states of a spinless particle with mass $m = 100$ subject to these potentials. (b) Use the TISE program to compute the eigenstates and eigenvalues for the potential $V_-(x)$, in the interval $x \in [0, R]$ with $R = 100$, with zero boundary conditions. (c) Prepare a two-slides presentation outlining the main features of this system. The slides must include a plot of the potentials, superposed to the energy levels you found (up to the height of the barrier), a plot of the bound eigenstates (up to three) of the Hamiltonian, and a plot for those states above the dissociation limit and below the potential barrier, for which the density of states per unit of energy appears to cluster (one plot per function).

Part 2: Identify a dissociative resonance in the potential, i.e., a state with energy above the dissociation threshold with a strong localized component within the barrier. To localize the function, switch off the B-spline coefficients with a smooth step function centered on the barrier, at x_s , and with width σ_s comparable to the barrier

$$|\psi\rangle = |\mathbf{B}\rangle \mathbf{q} \mapsto |\tilde{\psi}\rangle = |\mathbf{B}\rangle \tilde{\mathbf{q}}, \quad \tilde{q}_i = f_i q_i, \quad f_i = [1 - \text{erf}\left(\frac{i - i_0}{\sigma_0}\right)]/2$$

where $i_0 = x_s(N_{\text{nodes}} - 1)/R + k/2 - 1$ and $\sigma_0 \simeq \sigma_s(N_{\text{nodes}} - 1)/R$. Subsequently, you need to expand the resulting state in the basis of eigenstates of the potential,

$$\langle \varphi | \tilde{\psi} \rangle = \langle \varphi | \mathbf{B} \rangle \tilde{\mathbf{q}} = \mathbf{C}^\dagger \mathbf{S} \tilde{\mathbf{q}}.$$

At this point, it is easy to normalize the wavefunction as

$$|\tilde{\psi}\rangle \mapsto |\bar{\psi}\rangle = |\tilde{\psi}\rangle / \|\langle \varphi | \tilde{\psi} \rangle\|, \quad \langle \varphi | \bar{\psi} \rangle = \langle \varphi | \tilde{\psi} \rangle / \|\langle \varphi | \tilde{\psi} \rangle\|$$

Finally, compute the probability of finding the system in the original localized state $|\bar{\psi}\rangle$ as a function of time

$$P(t) = |\langle \bar{\psi} | \psi(t) \rangle|^2 = \left| \langle \bar{\psi} | e^{-i\hat{H}t/\hbar} | \bar{\psi} \rangle \right|^2 = \left| \langle \bar{\psi} | \varphi \rangle e^{-i\mathbf{E}t/\hbar} \langle \varphi | \bar{\psi} \rangle \right|^2,$$

plot it, and interpret its characteristic features.

5. **Hydrogen atom.** [Numerical, TISE code] The bound states of the hydrogen atom. $\psi_{n\ell m}(\vec{r})$ can be written in spherical coordinates as $\psi_{n\ell m}(\vec{r}) = \frac{u_{n\ell}(r)}{r} Y_{\ell m}(\theta, \phi)$, where $Y_{\ell m}$ are spherical harmonics, and the reduced radial function $u_{n\ell}$ satisfies the equation

$$-\frac{1}{2} u_{n\ell}''(r) + V_\ell(r) u_{n\ell}(r) = E_{n\ell} u_{n\ell}(r), \quad \text{where} \quad V_\ell(r) = \frac{\ell(\ell+1)}{2r^2} - \frac{1}{r}.$$

This equation requires that $u_{n\ell} \in L^2([0, \infty])$ and that $u_{n\ell}$ and its first ℓ derivatives vanish at the origin, $\left. \frac{d^k u_{n\ell}(r)}{dr^k} \right|_{r=0} = 0$ for $k = 0, 1, \dots, \ell$. The states are conventionally numbered $n = \ell + 1, \ell + 2, \dots$

Part 1: (a) Use the TISE program to solve this problem for $\ell = 0$ and $\ell = 1$ in the interval $r \in [0, R]$, with $R = 20$, with the assigned boundary conditions at the origin, and with zero boundary conditions at $x = R$. Tabulate the energies of the bound states up to $n = 4$ and compare them with the analytical value $E_n = -\frac{1}{2n^2}$. (b) Prepare a two-slides presentation identifying the main features of this system.

The slides must include a plot of the potentials $V_\ell(r)$ superposed to the energy levels you found, as well as a plot of $u_{n\ell}$ for $\ell \leq 1$ and $n \leq 3$ (one plot per function).

Part 2: Follow the same procedure described for problem N. 3 in the case of the orbital angular momentum $\ell = 1$.

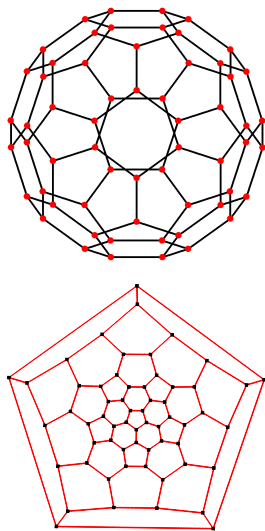
6. **Rigid rotor.** [*Numerical, script*] Consider the hindered rigid rotor with Hamiltonian

$$\hat{H} = -\frac{1}{2I} \frac{d^2}{d\theta^2} + V_0 \sin^2(3\theta/2), \quad \theta \in [0, 2\pi], \quad V_0 = 1, \quad I = 20.$$

Part 1: (a) Write a program (e.g., in Python) to compute the eigenstates and eigenvalues of this system in the basis of functions $\{\varphi_n(\theta) = e^{in\theta}/\sqrt{2\pi}, n = -N, -N+1, \dots, N-1, N\}$, with $N = 3$, $N = 10$, $N = 20$, and $N = 100$ [This is likely the simplest avenue, if you are familiar with Python. Alternatively, you can decide to use the TISE program, but you must pay special attention to enforce periodic boundary conditions. Ask the instructor]. Plot the energies of the first six eigenvalues as a function of N . (b) Prepare a two-slides presentation identifying the main features of this system. The slides must include a plot of the potential superposed to the energy levels you found, as well as a plot of the first six eigenfunctions.

Part 2: Find a linear combination of the first three eigenstates of the Hamiltonian that is strongly localized in one of the potential well, and plot its evolution with time. How long does it take to the particle to transition to the nearby wells? Repeat the same for the states 4, 5, and 6 and, if they are below the barrier, for states 7, 8, and 9. Plot the time needed for the function to drop at half its value in the original potential as a function of the energy gap between the expectation value of the initial function and the top of the potential barrier (as in P#2).

7. **Fullerene.** [*Geometrical & Numerical, script*]



Fullerene, a molecule with formula C_{60} , is an allotropic form of carbon. In C_{60} , all carbon atoms are equivalent, hybridized sp^2 and arranged on a sphere at the corners of a truncated icosahedron (soccer ball), with the residual $2p$ orbital with nodal plane tangent to the sphere. The diagrams below show fullerene in 3D as well as a 2D projection (Schlegel) illustrating how the atoms are connected. It is possible to approximate the energies of an electron in the delocalized π system of the molecule by limiting the attention to the space spanned by the 60 radial $2p$ orbitals. The effective Hamiltonian within this space can be assumed to be $\langle p_i | H | p_i \rangle = \alpha$ on the diagonal, $\langle p_i | H | p_j \rangle = \beta$ for all the pairs of nearest neighbors, and $\langle p_i | H | p_j \rangle = 0$ otherwise. Assume also $\langle p_i | p_i \rangle = 1$, and $\langle p_i | p_{j \neq i} \rangle = 0$.

Part 1: (a) Based on the connectivity of fullerene, use a programming language of your choice to build the Hamiltonian matrix $\mathbf{H}$ for the molecular π system, assuming $\alpha = 0$ and $\beta = -1$, and determine its eigenvalues and eigenvectors.

(b) Prepare a two-slides presentation identifying the main features of this system. The slides must include a plot of the eigenvalues for each of the eigenstates, in order of increasing energy, with a specification of the degeneracies that you find.

Part 2: Determine the expansion coefficients of a state localized on the carbon atom N.1 on the basis of eigenstates of the Hamiltonian that you have computed and calculate the subsequent evolution of the wavepacket. Plot the square module of the coefficient on atom 1 as a function of time and interpret the result.

8. **Infinite reflections.** *[Analytical & numerical, script]* Consider a particle with mass m in a box, $x \in [0, a]$, described at time $t = 0$ by a minimal uncertainty Gaussian wavepacket with central position x_0 , standard deviation $\sigma \ll x_0$, $\sigma \ll a - x_0$, and central momentum $\hbar k_0$.

Part 1: (a) Expand the function in the eigenstates of the box. Write a script to plot $|\psi(x, t)|^2$ as a function of time for $t \in [0, 10ma/\hbar k_0]$. (b) Prepare a two-slides presentation identifying the main features of this system. The slides must include a plot of the Gaussian wavepacket at various stages of its temporal evolution.

Part 2: Show that the same dynamics is reproduced, in the interval $[0, a]$, by the free evolution of the (non normalizable) wavefunction described at $t=0$ by

$$\Psi(x, 0) = \sum_{n \in \mathbb{Z}} [\psi(x - 2na, 0) - \psi(-x - 2na, 0)].$$

9. **Spontaneous decay.** *[Analytical & numerical, script]* A system has a Hamiltonian partitioned as $H = H_0 + V$, where H_0 is a self-adjoint operator with the following known eigenvalues and eigenvectors, $H_0|a\rangle = 0$, $H_0|n\rangle = |n\rangle n\delta$, $n = -N, -N + 1, \dots, N - 1, N$. On the basis of the eigenstates of H_0 , the operator V has the following non-zero matrix elements: $\forall n \langle a|V|n\rangle = V_0$, whereas all the other matrix elements are zero, $\langle a|V|a\rangle = \langle n|V|n'\rangle = 0$.

Part 1: (a) Write a program to compute the eigenvectors $|\psi_n\rangle$ and eigenvalues E_n of H for $V_0 = 1$, $\delta = 0.4$, and $N = 100$. Plot $\Delta\rho_n = \frac{1}{E_{n+1} - E_{n-1}} - \delta^{-1}$ as a function of E_n . (b) Prepare a two-slides presentation identifying the main features of this system. The slides must include a plot of the $\Delta\rho_n$ together with a fit of it with a function of the form $F(E) = A/(E^2 + \Gamma^2/4)$.

Part 2: Simulate the dynamics of the state $|\psi(t)\rangle$ with the initial condition $|\psi(0)\rangle = |a\rangle$. Compute the spectrum of this state, evaluate the probability of the system remaining in state $|a\rangle$ as a function of time,

$$P_a(t) = |\langle a|\psi(t)\rangle|^2 = \left| \langle a|e^{-i\hat{H}t/\hbar}|a\rangle \right|^2 = \left| \langle a|\varphi\rangle e^{-i\mathbf{E}t/\hbar} \langle \varphi|a\rangle \right|^2$$

and explain your observations.